

EDUCATION	The Chinese University of Hong Kong <i>Ph.D. in Computer Science and Engineering</i> • Advisor: Prof. Hong Xu • Research area: Machine Learning Systems	Hong Kong SAR, China 2022.08 - 2026.06 (<i>expected</i>)
	Northwestern Polytechnical University <i>B.E. in Computer Science and Technology</i> • GPA: 93.37/100, Rank: 1/247.	Xi'an, China 2018.09 - 2022.06
RESEARCH INTEREST	I am broadly interested in Machine Learning Systems (MLSys), including the following topics: 1. Resource Scheduling: Maximizing the allocation and utilization of available resources to accommodate workloads in the cluster. 2. ML Inference: Designing effective and scalable systems for efficiently serving large volumes of requests. 3. Distributed Training: Developing and optimizing strategies for efficient training of large models at scale.	
PUBLICATIONS	1. Xin Tan, Yicheng Feng, Yu Zhou, Yimin Jiang, Yibo Zhu, Hong Xu, Dynamic Compute and Network Orchestration for Disaggregated RL. <i>ACM Special Interest Group on Data Communication (SIGCOMM)</i>, 2026. 2. Xin Tan, Yuetao Chen, Yimin Jiang, Xing Chen, Kun Yan, Nan Duan, Yibo Zhu, Daxin Jiang, Hong Xu, DSV: Exploiting Dynamic Sparsity to Accelerate Large-Scale Video DiT Training. <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i>, 2026. 3. Xin Tan, Yimin Jiang, Yitao Yang, Hong Xu, Towards End-to-End Optimization of LLM-based Applications with Ayo. <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i>, 2025. 4. Xin Tan, Yicheng Feng, Yu Zhou, Yimin Jiang, Yibo Zhu, Hong Xu, RFabric: A Reconfigurable Network for the Rhythms of Disaggregated RL. <i>The 1st Frontier AI Systems Workshop (FAISys)</i>, 2025. 5. Xin Tan, Jiamin Li, Yitao Yang, Jingzong Li, Hong Xu, Arlo: Serving Transformer-based Language Models with Dynamic Input Lengths. <i>ACM International Conference on Parallel Processing (ICPP)</i>, 2024. 6. Kaihua Liang, Xin Tan, An Zhong, Hong Xu, Marco Canini, FOCUS: DLLMs Know How to Tame Their Compute Bound. <i>International Conference on Machine Learning (ICML)</i>, 2026. 7. Kaiwen Chen, Xin Tan, Minchen Yu, Jingzong Li, Hong Xu, MemShare: Memory Efficient Inference for Large Reasoning Models through KV Cache Reuse. <i>IEEE/ACM International Symposium on Quality of Service (IWQoS)</i>, 2026. 8. Yicheng Feng, Xin Tan, Kin Hang Sew, Yimin Jiang, Yibo Zhu, Hong Xu, Frontier: Simulating the Next Generation of LLM Inference Systems. <i>SOSP Workshop on Practical Adoption Challenges of ML for Systems (SOSP-PACMI)</i>, 2025.	

INTERNSHIPS	System Group, StepFun Beijing, China 2024.08 - 2026.01 - Identified attention bottlenecks in video DiTs with long video inputs and analyzed dynamic sparse attention patterns. Co-designed algorithms and systems to scale video-DiT sparse training, implementing the solution to achieve a 3.02x throughput improvement on 128 GPUs. - Explored next-generation data center designs for LLM post-training infrastructure.
	Network Research Group, Microsoft Research Asia Remote 2021.10 - 2022.03 - Developed a real-time, non-intrusive monitoring system to collect critical AI infrastructure metrics (GPU utilization, network/NVLink bandwidth), and designed a scalable analytics tool to evaluate ML workloads across large-scale clusters. - Analyzed six months of datacenter workload data to characterize resource usage and network patterns of various AI tasks, providing actionable recommendations to optimize cloud infrastructure and software stack.
AWARDS AND HONORS	Student Travel Grant, ASPLOS 2025 2025.04 Full Postgraduate Scholarship, The Chinese University of Hong Kong 2022.08 - 2026.06 Outstanding Graduate, Northwestern Polytechnical University 2022 National Scholarship, Ministry of Education (China) 2020 National Scholarship, Ministry of Education (China) 2019 Champion, International Underwater Robot Competition 2020
SKILLS	Languages: Chinese, English Programming: Python, C/C++, CUDA, PyTorch, Triton Systems: Megatron-LM, Ray
ACADEMIC SERVICES	Reviewer: <i>IEEE Transactions on Network Science and Engineering</i> Shadow Program Committee: <i>ACM EuroSys 2026</i> Artifact Evaluation Committee: <i>USENIX OSDI/ATC 2025, USENIX OSDI/ATC 2024, USENIX FAST 2025, ACM CoNEXT 2025, ACM EuroSys 2025 Spring/Fall</i>
TEACHING	Teaching Assistant: <i>CSCI 3150, Introduction to Operating Systems, CUHK. 2023 Spring, CSCI 1120, Introduction to Computing Using C++, CUHK. 2022 Fall.</i>